Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

MIT MPL - B.Tech. V Semester Aeronautical/ Automobile Engineering - End Semester Examination - Nov-Dec 2024

FINITE ELEMENT METHODS [AAE 3125]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

* Answer all questions.
* Use of Course Data Sheet is allowed.
* Assume the missing data suitably.
* Write neatly and legibly.

1) Calculate the unknown displacements for the spring assembly given in Figure 1. (4)

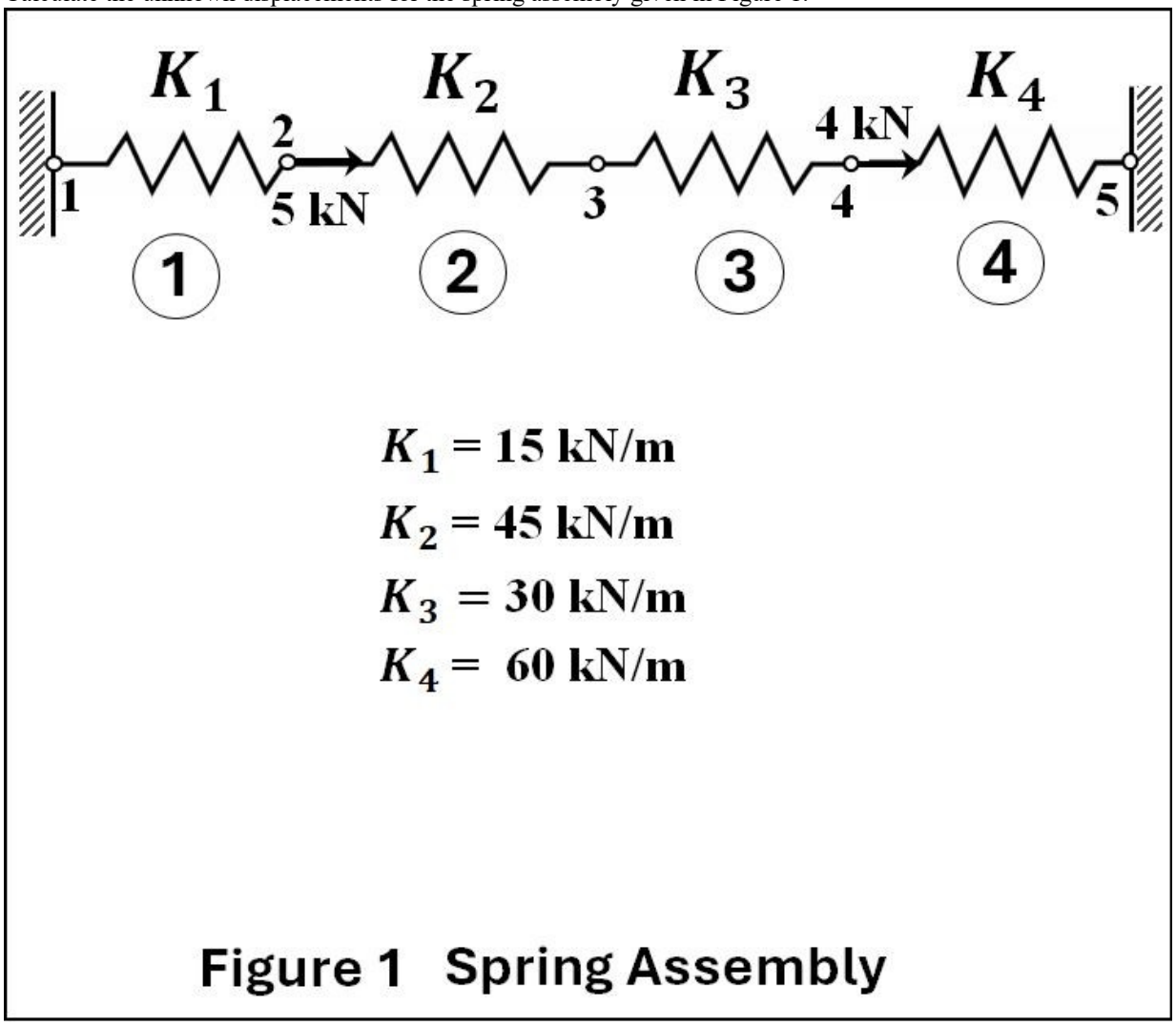


Figure 1 Spring Assembly

2) For the spring assembly shown in Figure 1, determine the reactions at the supports. (2)

3) A bar is subjected to surface traction of $4x^2$ (in kN), as shown in Figure 2. Determine the nodal displacements considering two equal-length elements. Let $A=6\times10^{-4}$ m^2, E=200 GPa. (4)

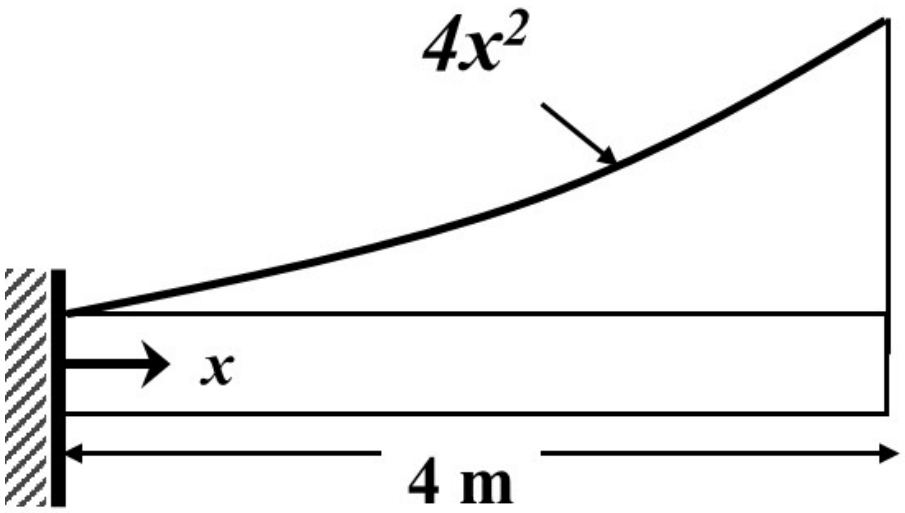


Figure 2

4) For the plane truss shown in Figure 3, determine the horizontal and vertical displacements of node 1. (4)

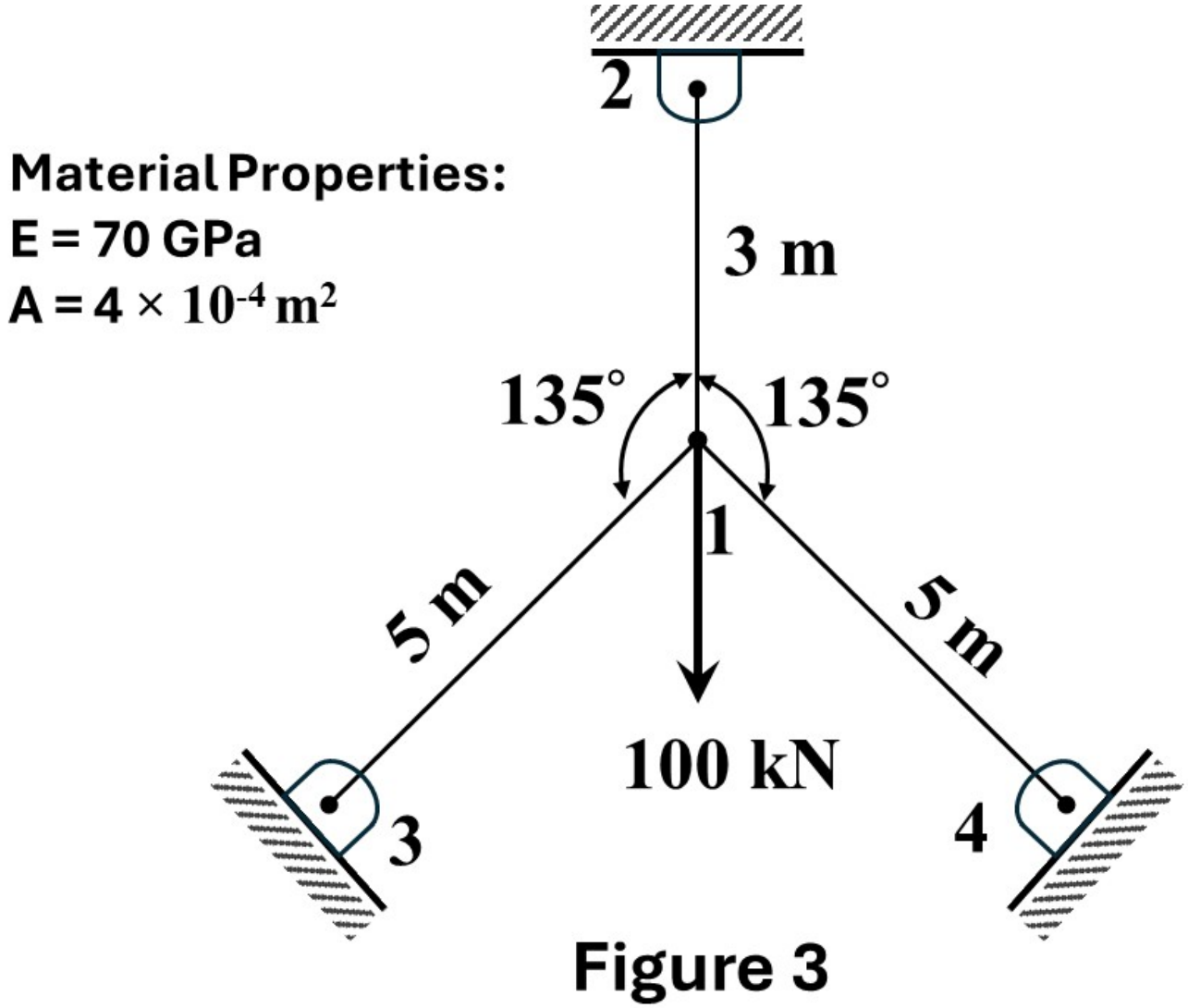


Figure 3

5) For the plane truss shown in Figure 3, calculate the reactions at the supports. (3)

6) For the plane truss shown in Figure 3, determine the stresses in each element. (3)

7) For the fixed-fixed beam assembly shown in Figure 4, derive the global stiffness matrix and calculate the nodal forces at the mid-span due to the distributed load. (4)

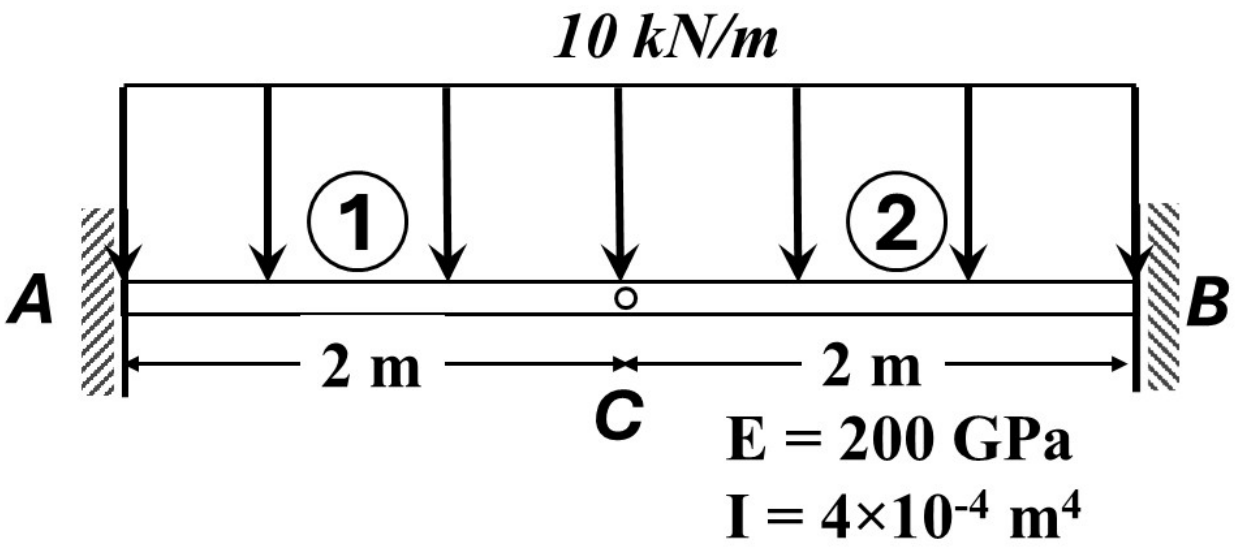


Figure 4

8) For the beam assembly in Figure 4, determine mid-span deflection and rotation. (3)

9) For the beam assembly in Figure 4, determine the reactions at the supports. (3)

10) A discretized 2D Plane Stress domain shown in Figure 5 and determine the stiffness matrix for Element (1). Given E= 200 GPa, υ=0.3, t=0.04 m. (4)

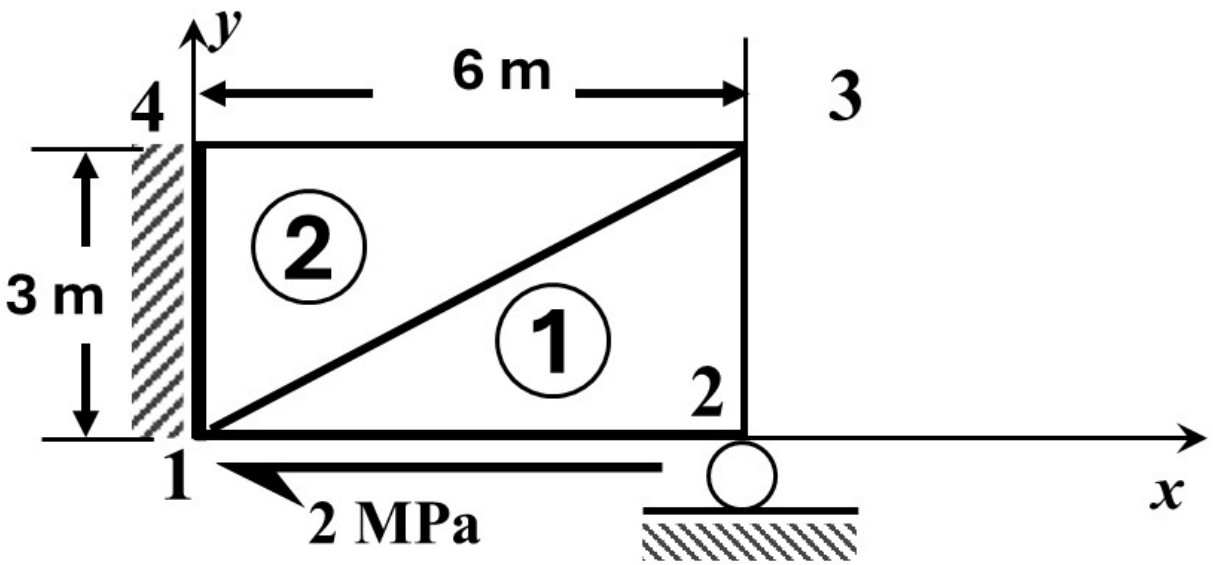


Figure 5

11) For the 2D plane stress domain shown in Figure 5, evaluate the stiffness matrix for Element (2). (3)

12) For the 2D Plane Stress problem shown in Figure 5, establish the force-displacement relation in terms of the global stiffness matrix. (3)

13) For the 2D Plane Stress problem in Figure 5, determine the unknown displacements. (4)

14) For the 2D Plane Stress problem in Figure 5, find out the reactions at the supports. (3)

15) For the triangular element shown in Figure 6, determine the values of the point function coefficients, 'α_i', 'β_i' and 'γ_i' at each node, respectively. (3)

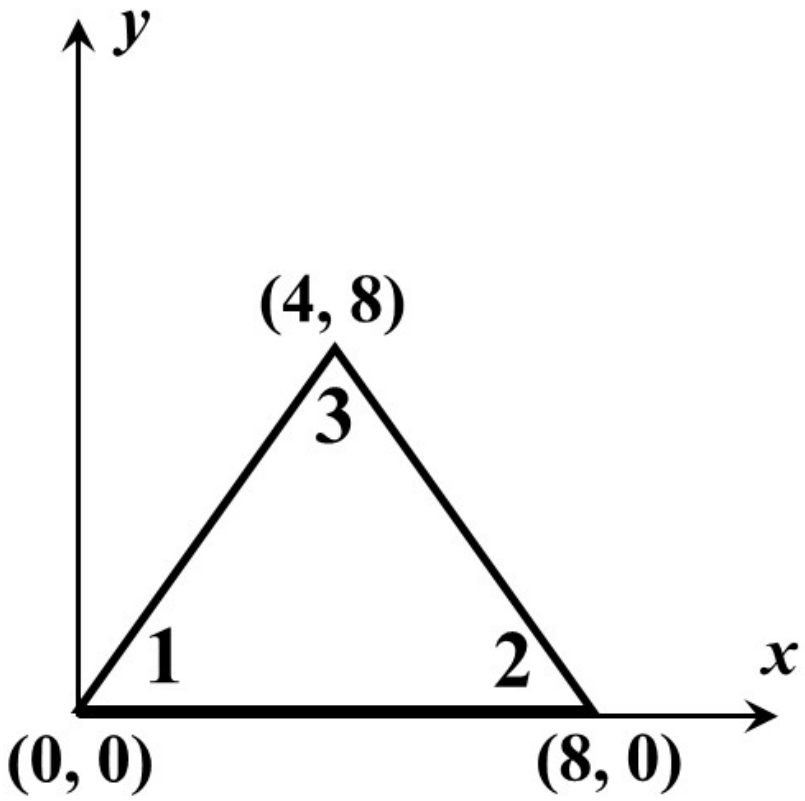


Figure 6

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